For

$\because$ X is uniform between -1 to 1

$\Rightarrow f_X(x) = \frac{1}{2}$

$\therefore$ Mean :

$$E[X] = \int_{-1}^{1} x \cdot \frac{1}{2} dx = 0$$

$$E[X^2] = \int_{-1}^{1} x^2 \frac{1}{2} dx = \frac{1}{3}$$

$$E[X^3] = \int_{-1}^{1} x^3 \frac{1}{2} dx = 0$$

$\therefore E[N] = 0$ and $E[N^2] = 1$

$(\because$ mean $= 0)$ $(\because$ Variance $= 1)$

Also $Y = X + N$

The covariance is

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$

$\Rightarrow \because E[Y] = E[X] + E[N] = 0$

Now,

$$E[XY] = E[X(X+N)]$$

$$= E[X^2] + E[XN]$$

$\because$ X and N are independent

$\Rightarrow E[XN] = E(X)E(N) = 0$

$\therefore E[XY] = E[X^2] = \frac{1}{3}$

$\Rightarrow Cov(X, Y) = \frac{1}{3}$ $(\because E[X] = 0)$

Also $Z = X^2 + N$

$$E[Z] = E[X^2] + E[N] = \frac{1}{3}$$

Now,

$$E[XZ] = E[X(X^2 + N)]$$

$$= E[X^3] + E[XN]$$

$E[X^3] \to 0$, $E[XN] \to 0$

$$E[XZ] = 0$$

$$\Rightarrow Cov(X, Z) = 0 \quad (\because E[X] = 0)$$